



Naba Jakirhusain Hosmani

Electronics & Communication Engineering

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 github.com/naba-husain naba-portfolio-psi.vercel.app Indian

EDUCATION

Bachelor of Engineering (B.E.) in Electronics and Communication Engineering

KLE Technological University
CGPA : 8.48

2023 – 2027
Hubballi, India

Pre University in Science

Chetan PU Science College

2020 – 2022
Hubballi, India

Class X

Smt. Nirmala K Thakkar Eng Med School

2020
Hubballi, India

SKILLS

Programming Languages

Python, Java (Fundamentals)

AI & Machine Learning

Data Preprocessing, Feature Engineering,
Supervised Learning, Model Evaluation

Computer Vision

OpenCV, YOLO (v5, v8, v11), Image Processing
Image Classification, Object Detection

Tools & Technologies

Git, GitHub, VS Code, Jupyter Notebook,
Google Colab, Kaggle, MATLAB, Simulink, Ollama

Core Concepts

Data Structures & Algorithms (DSA),
Object Oriented Programming (OOP),
Operating System, Computer Networks

Generative AI

Large Language Models (LLMs),
Retrieval-Augmented Generation (RAG),
Prompt Engineering

Soft Skills

Analytical Thinking, Problem Solving,
Communication, Teamwork, Adaptability, Time
Management

CERTIFICATES

AI Foundations Associate

Oracle

Junos, Associate (JNCIA-Junos)

HPE Juniper

Generative AI

Oracle

Data Science

Oracle

MATLAB Onramp

MathWorks

Signal Processing Onramp

MathWorks

LANGUAGES

- English
- Hindi
- Kannada
- Urdu

AWARDS

First Place – Technical Poster Presentation

INSIGNIA, SDM College of Engineering & Technology

Presented an engineering project and secured first place among participating teams.

First Place – PCB Design Competition

Department of Electronics & Communication Engineering, SDMCET

Secured first place in a PCB design competition demonstrating circuit design and implementation skills.

Smart India Hackathon 2025 – Internal Round Selection

KLE Technological University

Selected as Team ResQ in the Internal Smart India Hackathon 2025 for developing an AI-driven earthquake-stabilized dialysis system.

PROJECTS

RAG-Based Video Question Answering with Timestamp Localization

Python | Whisper | Ollama (Llama 3.2) | BGE-m3 | RAG | NLP

- Developed a RAG-based pipeline to answer questions from lecture videos using Whisper transcription, semantic embeddings, and retrieval.
- Implemented timestamp localization with BGE-m3 embeddings and cosine similarity to retrieve relevant video segments.
- Evaluated the system achieving **90% Top-1 retrieval accuracy**; research paper accepted at **INCECT 2026**.

Multi-Robot Deadlock Detection & Physics-Aware Dataset Generation

ROS2 (Humble) | Gazebo | Python | Nav2

- Built a ROS2 and Gazebo-based framework for multi-robot deadlock detection and labeled dataset generation.
- Simulated corridor, T-junction, and cross-junction scenarios to generate **10,000+ labeled navigation samples**.
- Implemented rule-based deadlock detection and resolution strategies for autonomous mobile robots (AMRs).

AI Powered HR Grievance Management System

Python | NLP | RAG

- Collaborated on developing an AI-powered HR grievance management system.
- Assisted in implementing NLP and RAG for grievance categorization and query resolution.
- Contributed to testing and improving the system's functionality.

Earthquake-Stabilized Dialysis System

Python | Machine Learning | Time-Series Analysis | Disaster Management

- Developed the machine learning module to analyze real-time vibration sensor data for earthquake risk prediction.
- Built a predictive model to trigger automated safety actions for dialysis machines during seismic events.
- Selected for the **Smart India Hackathon 2025 Internal Round** under the Disaster Management theme.

Helmet Detection System

Python | OpenCV | YOLOv8 | Computer Vision

- Developed a real-time helmet detection system for riders and pillion passengers using the YOLOv8 object detection framework.
- Applied image preprocessing and dataset augmentation techniques to improve model robustness across varying traffic conditions.
- Achieved **92.3% mAP@0.5** on the validation dataset.

Smart Accident Detection and Alert System

Arduino | Embedded Systems | Accelerometer | Bluetooth

- Developed an Arduino-based accident detection system using a 3-axis accelerometer to detect high-impact collisions.
- Integrated Bluetooth communication to transmit accident alerts to a mobile application in real time.
- Designed and tested the complete hardware prototype, including sensor interfacing, alert logic, and power supply circuitry.

PUBLICATIONS

RAG-Based Video Question Answering with Timestamp Localization

International Conference on Emerging Computing Technologies (INCECT 2026)

- Accepted for presentation at **INCECT 2026**.
- Proposed a **RAG-based video question answering framework** with timestamp localization.
- Achieved **90% Top-1 retrieval accuracy** on a benchmark of lecture videos.